

Final Project

Effect of Inlet Mach Number on Stagnation Pressure in a Wind Tunnel with a Forward-Facing Step

COE 347: Intro to Computational Fluid Dynamics

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1 Introduction

This report extends OpenFOAM Assignment 3 by investigating how inlet Mach number affects the stagnation pressure at the foot of the forward-facing step in a two-dimensional supersonic wind tunnel. The central question is whether `rhoCentralFoam` correctly predicts how the stagnation pressure at $(x, y) = (0.6, 0)$ scales with Mach number, and how well the numerical results agree with the analytical normal shock prediction across $M = 1.5, 2, 3, 4$, and 5. The $M = 3$ case carries over from Assignment 3 as the baseline; all other parameters are unchanged and only the inlet velocity is varied. The analytical reference at each Mach number is computed using the two-step normal shock and isentropic stagnation procedure described in Section 3, and the $M = 3$ results are validated against the benchmark solution of Woodward and Colella [1].

2 Governing Equations

The flow is governed by the two-dimensional compressible Euler equations for a calorically perfect ideal gas with $\gamma = 1.4$:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = \mathbf{0}, \quad \mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho u H \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho v H \end{bmatrix} \quad (1)$$

where $H = E + p/\rho$ is the specific total enthalpy, $E = e + \frac{1}{2}(u^2 + v^2)$ is the specific total energy, and the calorically perfect equation of state closes the system: $p = (\gamma - 1)[\rho E - \frac{1}{2}\rho(u^2 + v^2)]$, with $e = p/[\rho(\gamma - 1)]$ and $T = p/(\rho R)$. The local sound speed and Mach number are $a = \sqrt{\gamma p/\rho}$ and $M = |\mathbf{u}|/a$.

3 Analytical Stagnation Pressure

The stagnation pressure at the step is estimated analytically by combining normal shock jump relations with isentropic flow. The incoming supersonic flow passes through a normal shock at the step face at $x = 0.6$ m, and the subsonic post-shock flow is then brought isentropically to rest at $(0.6, 0)$.

Step 1 — Normal shock jump relations:

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1}, \quad M_2^2 = \frac{(\gamma - 1)M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} \quad (2)$$

Step 2 — Isentropic deceleration to stagnation:

$$\frac{p_0}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_2^2\right)^{\gamma/(\gamma - 1)} \quad (3)$$

For the $M = 3$ baseline ($p_1 = 1$ Pa, $\gamma = 1.4$): Step 1 gives $p_2 = 10.333$ Pa and $M_2 = 0.4752$; Step 2 gives $p_0 = 1.1672 \times 10.333 \approx 12.06$ Pa. For reference, the isentropic (no-shock) total pressure would be $p_{0,\text{isen}} = 36.73$ Pa, so the normal shock reduces stagnation pressure by a factor of 0.328 — a substantial total pressure loss that grows monotonically with M_1 . Table 1 summarizes the results at all five Mach numbers alongside the fine-mesh numerical values.

Table 1: Analytical intermediate values and stagnation pressures at each inlet Mach number ($\gamma = 1.4$, $p_1 = 1$ Pa). Numerical values probed at (0.6, 0.005, 0) on the fine mesh at $T = 4$ s.

M_1	p_2/p_1	M_2	p_0/p_2	$p_{0,\text{anal.}}$ [Pa]	$p_{0,\text{num.}}$ [Pa]
1.5	2.4583	0.7011	1.3885	3.4133	3.9654
2.0	4.5000	0.5774	1.2534	5.6404	6.0208
3.0	10.3333	0.4752	1.1672	12.0610	12.0440
4.0	18.5000	0.4350	1.1388	21.0681	20.9771
5.0	29.0000	0.4152	1.1260	32.6535	32.5075

4 Computational Setup

All cases use OpenFOAM v7 with `rhoCentralFoam` and the Kurganov–Tadmor central-upwind flux scheme [3] with van Leer reconstruction for ρ , $\mathbf{U}$, and T . This scheme belongs to the class of high-resolution central methods evaluated by Emery [2] for inviscid flow problems and is well-suited to shock-capturing without an explicit Riemann solver. Time integration uses the first-order explicit Euler scheme. The gas is inviscid and calorically perfect with $\gamma = 1.4$, normalized so $a = 1$ m/s at $T = 1$ K, meaning the inlet Mach number equals the inlet speed in m/s. Boundary conditions are listed in Table 2; mesh parameters in Table 3. All simulations run to $T = 4$ s with $Co_{\text{max}} = 0.2$.

Table 2: Boundary conditions (from 0/p, 0/T, 0/U).

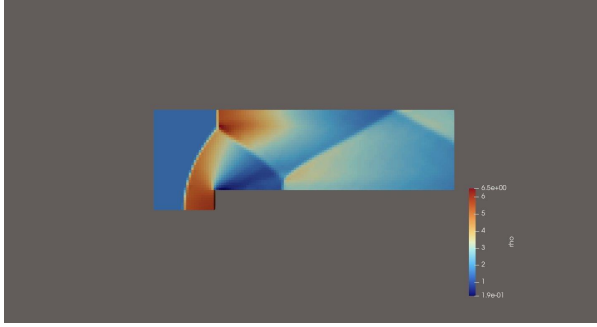
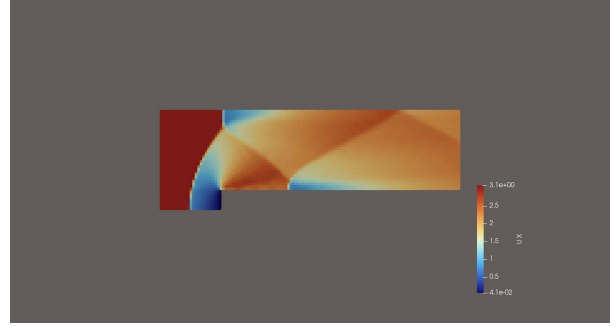
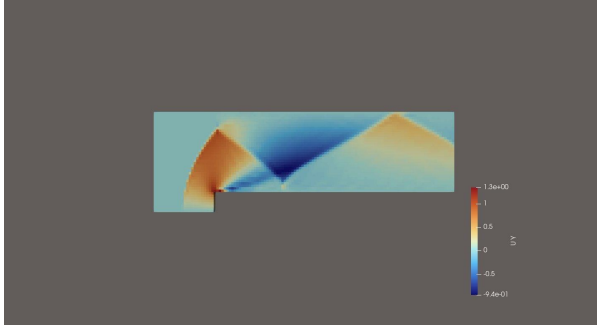
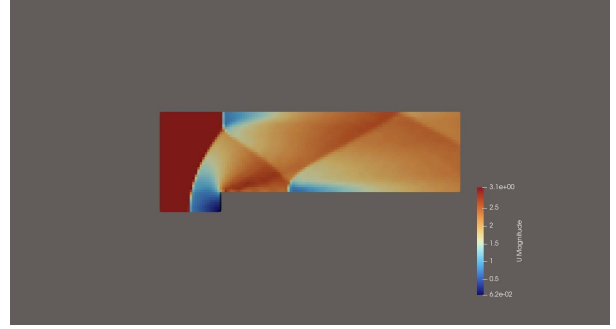
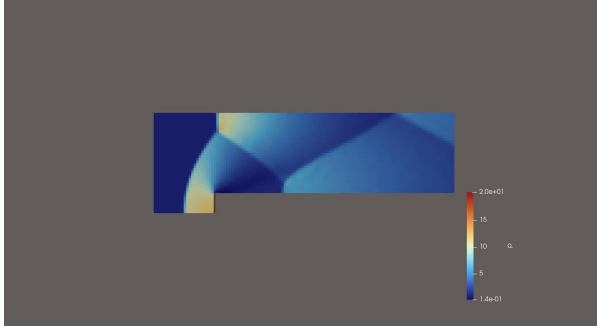
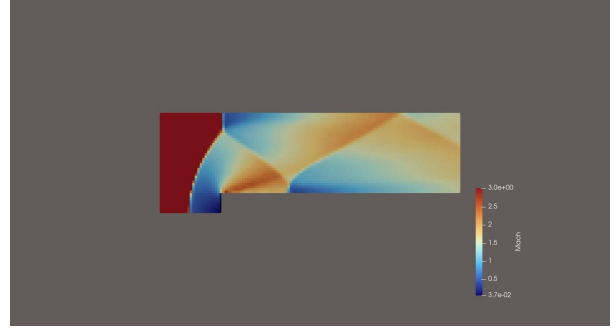
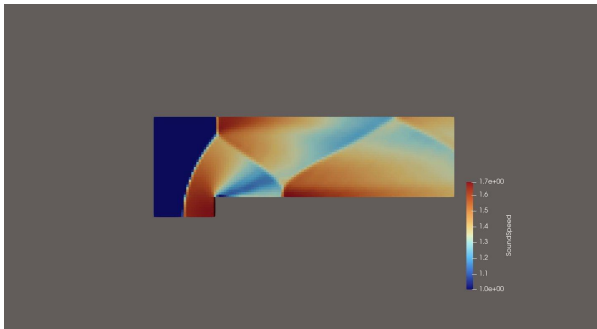
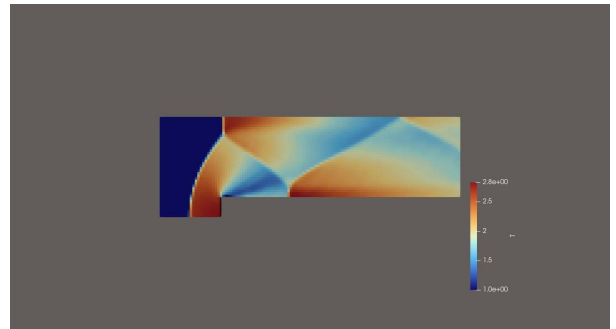
Patch	Type	p	T	U	Notes
inlet	patch	fixedValue=1 Pa	fixedValue=1 K	fixedValue ($M, 0, 0$) m/s	Varies per case
outlet	patch	zeroGradient	inletOutlet	inletOutlet	Supersonic outflow
top	symmetryPlane	symmetryPlane	symmetryPlane	symmetryPlane	Top tunnel wall
bottom	symmetryPlane	symmetryPlane	symmetryPlane	symmetryPlane	Inlet floor ($x < 0.6$ m)
obstacle	patch	zeroGradient	zeroGradient	slip	Step faces (inviscid)
defaultFaces	empty	empty	empty	empty	2D front/back

Table 3: Mesh parameters. Fine mesh used for all five Mach number cases; coarse and medium used only for the convergence study at $M = 1.5$ and $M = 5$.

Mesh	Total Cells	$\Delta x = \Delta y$ [m]	Δt (approx.) [s]	Cases
Coarse	4,032	0.025	0.00125	$M = 1.5, 5$
Medium	16,128	0.0125	0.000625	$M = 1.5, 5$
Fine	64,512	0.00625	0.000313	All five

5 $M = 3$ Baseline Flow Field

Figure 1 shows the eight flow field quantities for the $M = 3$ baseline on the fine mesh at $T = 4$ s. The solution captures all expected features of Mach 3 flow over a forward-facing step: a curved bow shock just upstream of the step at $x = 0.6$ m, a near-normal shock at the step face, an oblique shock propagating into the upper channel, a Mach reflection off the top wall with a well-defined Mach stem, and a stagnation region at $(0.6, 0)$. These features are compared quantitatively to Woodward and Colella [1] in Section 8.2.

(a) Density ρ [kg/m³](b) x -velocity u [m/s](c) y -velocity v [m/s](d) Velocity magnitude $|\mathbf{u}|$ [m/s](e) Pressure p [Pa](f) Mach number M (g) Sound speed a [m/s](h) Temperature T [K]Figure 1: $M = 3$ baseline flow field at $T = 4$ s, fine mesh.

6 Flow Fields Across Inlet Mach Numbers

Figure 2 shows pressure and Mach number fields on the fine mesh at $T = 4$ s for $M = 1.5, 2, 4$, and 5 . The progression is physically consistent across the full range. At $M = 1.5$ the shock is weak and diffuse — the pressure rise at the step face is modest and the Mach number field shows no sharp discontinuity, consistent with the near-sonic limit where shock jumps approach zero. By $M = 2$ the bow shock is clearly defined and the subsonic pocket behind the step is visible. At $M = 4$ and $M = 5$ the shock is thin and tightly curved, the stagnation pressure at the step is sharply elevated, and both the Mach reflection off the top wall and the slipline emanating from the triple point are well resolved. This monotonic sharpening of the shock structure with increasing M is consistent with the normal shock pressure ratio p_2/p_1 in Table 1, which grows from 2.46 at $M = 1.5$ to 29.0 at $M = 5$.

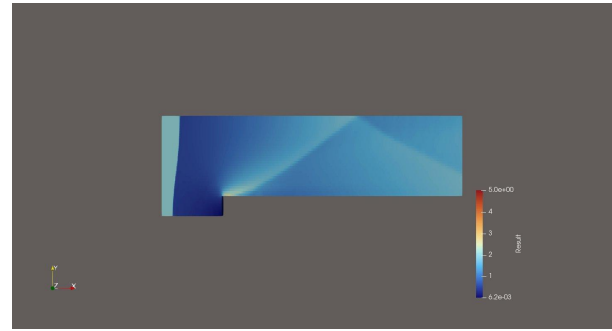
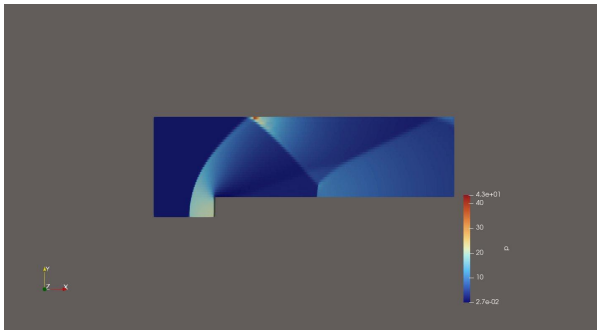
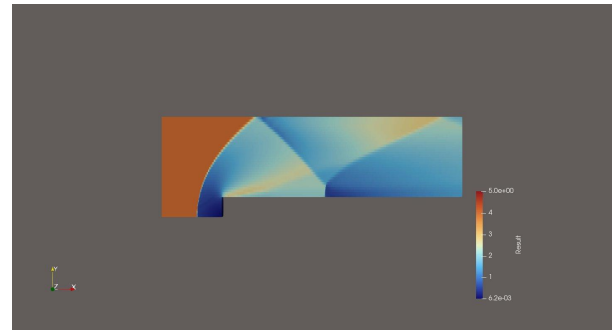
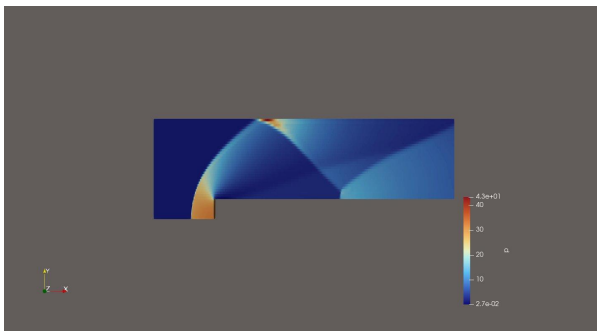
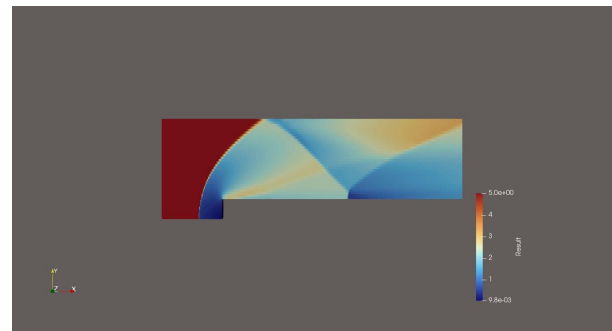
(a) Pressure, $M = 1.5$ (b) Mach number, $M = 1.5$ (c) Pressure, $M = 2$ (d) Mach number, $M = 2$ (e) Pressure, $M = 4$ (f) Mach number, $M = 4$ (g) Pressure, $M = 5$ (h) Mach number, $M = 5$

Figure 2: Pressure p [Pa] and Mach number fields on the fine mesh at $T = 4$ s for $M = 1.5$, 2, 4, and 5.

7 Mesh Convergence Study

A mesh refinement study is carried out at $M = 1.5$ and $M = 5$ to bound convergence behavior across the full Mach number range. These two cases represent the weakest and strongest shocks respectively, so demonstrating convergence at both extremes gives confidence in the fine-mesh results for the intermediate cases $M = 2, 3$, and 4 .

7.1 $M = 1.5$ — Weak Shock

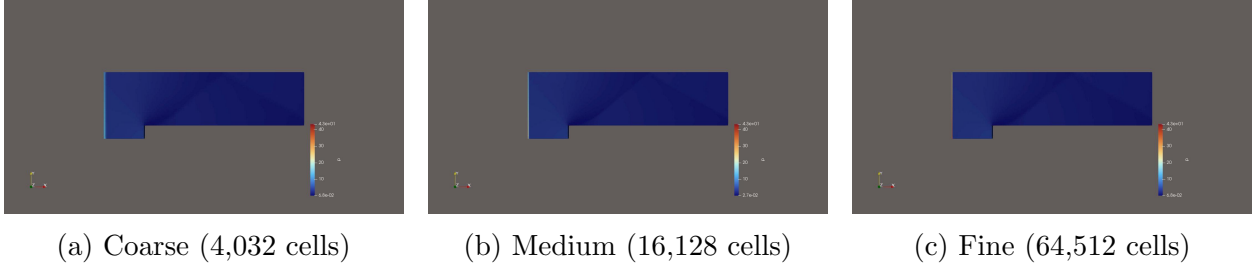


Figure 3: Pressure p [Pa] at $T = 4$ s — $M = 1.5$, three mesh levels.

Table 4: Mesh convergence of stagnation pressure at $(0.6, 0)$ for $M = 1.5$. Analytical value $p_0 = 3.4133$ Pa.

Mesh	Cells	$p_{0,\text{num.}}$ [Pa]	$p_{0,\text{anal.}}$ [Pa]	Error [%]
Coarse	4,032	3.9496	3.4133	15.7
Medium	16,128	3.9579	3.4133	15.9
Fine	64,512	3.9654	3.4133	16.2

The stagnation pressure at $M = 1.5$ changes by less than 0.5% across all three mesh levels, confirming the solution is fully mesh-converged. Yet all three meshes consistently overshoot the analytical value by $\sim 16\%$. Since no residual discretization error remains to extrapolate away, Richardson extrapolation is not applicable here. The persistent offset is instead a systematic modeling discrepancy from the breakdown of the normal shock assumption at low Mach numbers.

At $M_1 = 1.5$ the shock at the step is oblique rather than purely normal. The shock angle is approximately $\beta \approx 72^\circ$ as visible in Figure 2b, giving a normal Mach component of $M_{1n} = M_1 \sin \beta \approx 1.43$. Substituting into the normal shock relations (Equation 2) yields $p_2/p_1 \approx 2.21$ and $M_{2n} \approx 0.726$; applying Equation (3) then gives an oblique-shock stagnation pressure of:

$$p_{0,\text{oblique}} \approx 2.21 \times (1 + 0.2 \times 0.726^2)^{3.5} \approx 3.09 \text{ Pa} \quad (4)$$

This is closer to the numerical result of 3.97 Pa than the normal shock prediction of 3.41 Pa; the remaining gap is attributable to shock curvature and the finite distance of the probe

point from the exact stagnation point. The 16% error relative to the normal shock model is therefore a limitation of the analytical reference, not the solver.

7.2 $M = 5$ — Strong Shock

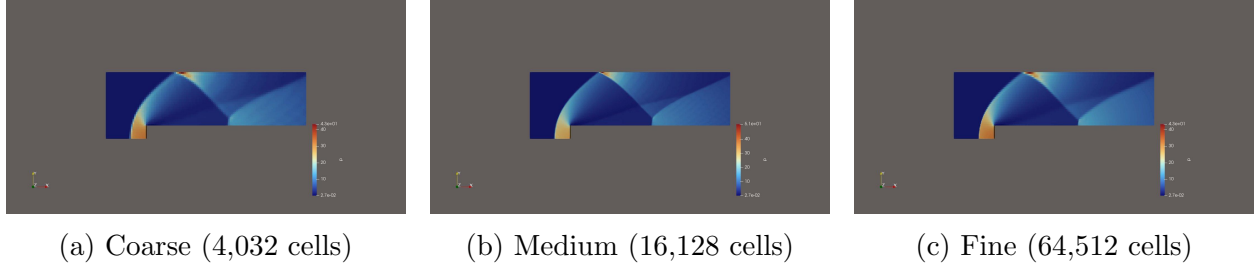


Figure 4: Pressure p [Pa] at $T = 4$ s — $M = 5$, three mesh levels.

Table 5: Mesh convergence of stagnation pressure at $(0.6, 0)$ for $M = 5$. Analytical value $p_0 = 32.6535$ Pa.

Mesh	Cells	$p_{0,\text{num.}}$ [Pa]	$p_{0,\text{anal.}}$ [Pa]	Error [%]
Coarse	4,032	32.0566	32.6535	1.83
Medium	16,128	32.3959	32.6535	0.79
Fine	64,512	32.5075	32.6535	0.45

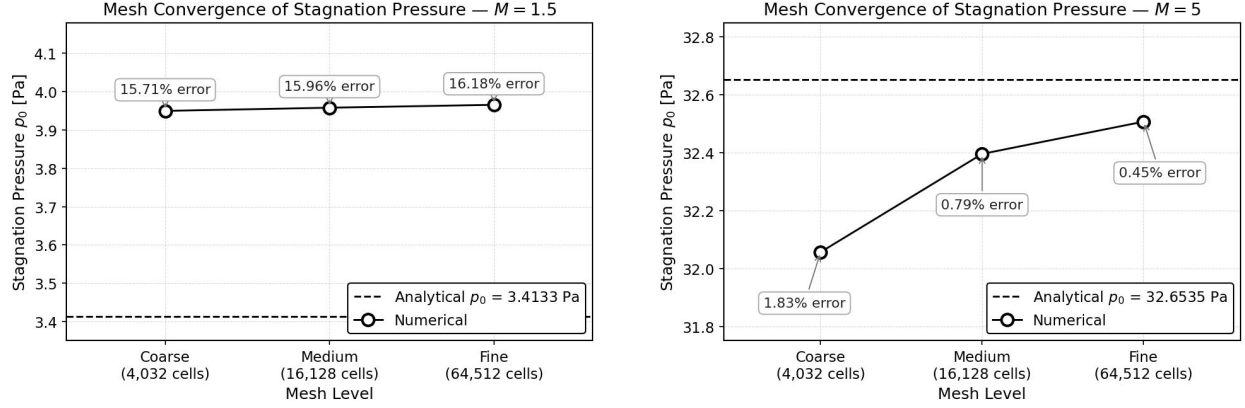
The $M = 5$ stagnation pressure converges monotonically toward the analytical value as the mesh is refined, with the error dropping from 1.83% on the coarse mesh to 0.45% on the fine mesh. This clean monotonic behavior permits a formal Richardson extrapolation. With refinement ratio $r = 2$ and three successive mesh levels f_1 (coarse), f_2 (medium), f_3 (fine), the observed order of convergence is:

$$p_{\text{obs}} = \frac{\ln\left(\frac{f_1 - f_2}{f_2 - f_3}\right)}{\ln r} = \frac{\ln(0.3393/0.1116)}{\ln 2} = \frac{\ln(3.041)}{0.6931} \approx 1.60 \quad (5)$$

The Richardson-extrapolated value is:

$$p_0^* = f_3 + \frac{f_3 - f_2}{r^{p_{\text{obs}}} - 1} = 32.5075 + \frac{0.1116}{2.031} \approx 32.563 \text{ Pa} \quad (0.28\% \text{ error}) \quad (6)$$

The extrapolated value of 32.563 Pa improves on the fine-mesh result (0.45% error) and agrees with the analytical value to within 0.28%. The observed order $p_{\text{obs}} \approx 1.6$ is consistent with the expected behavior of the Kurganov–Tadmor scheme with van Leer reconstruction: nominally second-order in smooth regions but locally first-order at the shock discontinuity, yielding an effective global rate between 1 and 2 on this shock-dominated domain.



(a) $M = 1.5$: converged at ~ 3.97 Pa, 16% above analytical (modeling error, not numerical).

(b) $M = 5$: monotonic convergence from 1.83% to 0.45%; Richardson extrapolation gives 0.28%.

Figure 5: Stagnation pressure at $(0.6, 0)$ vs. mesh level. Dashed line: analytical value.

8 Comparison with Analytical Results and Literature

8.1 Numerical vs. Analytical Stagnation Pressure

Table 6: Numerical vs. analytical stagnation pressure at $(0.6, 0)$ for all five Mach numbers, fine mesh at $T = 4$ s.

M_1	$p_{0,\text{anal.}}$ [Pa]	$p_{0,\text{num.}}$ [Pa]	Error [%]
1.5	3.4133	3.9654	16.2
2.0	5.6404	6.0208	6.75
3.0	12.0610	12.0440	0.14
4.0	21.0681	20.9771	0.43
5.0	32.6535	32.5075	0.45

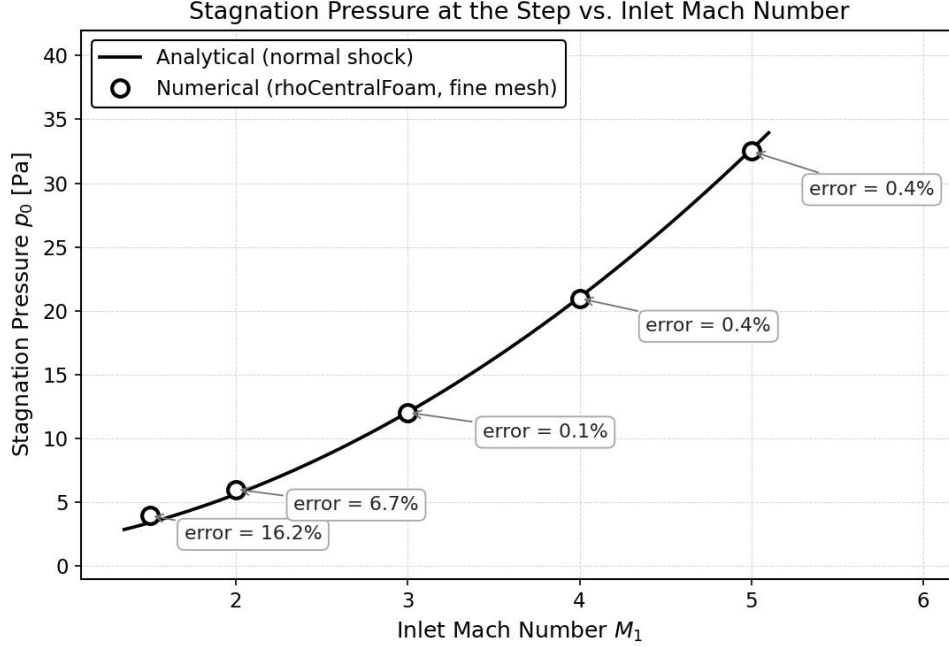


Figure 6: Stagnation pressure p_0 at $(0.6, 0)$ vs. inlet Mach number. Solid line: analytical normal shock prediction. Open circles: `rhoCentralFoam` fine-mesh results at $T = 4$ s. Percent errors annotated at each point.

The solver correctly captures the monotonic increase in stagnation pressure with Mach number, tracking the analytical curve closely for $M \geq 2$ (Table 6, Figure 6). Errors are below 0.5% for $M \geq 3$ and rise to 6.75% at $M = 2$ and 16.2% at $M = 1.5$. The $M = 1.5$ error is a modeling artifact established in Section 7.1: the solution is mesh-converged and the discrepancy traces quantitatively to the oblique shock geometry at the step, not any deficiency in the solver. Notably, the best agreement occurs at $M = 3$ rather than at the highest Mach number. This reflects two competing error sources: shock obliqueness inflates the analytical model error at low M , while numerical diffusion from the finite mesh and first-order time scheme inflates the numerical error at high M . At $M = 3$ these effects are both small and largely cancel, producing the best overall agreement.

8.2 Comparison with Woodward and Colella (1984)

The $M = 3$ results are compared against the benchmark solution of Woodward and Colella [1], who use the PPMLR scheme on a uniform grid with $\Delta x = \Delta y = 1/80$ and report density and pressure contours at $T = 4$ s (their Figure 3) along with a three-grid convergence study (their Figure 7).

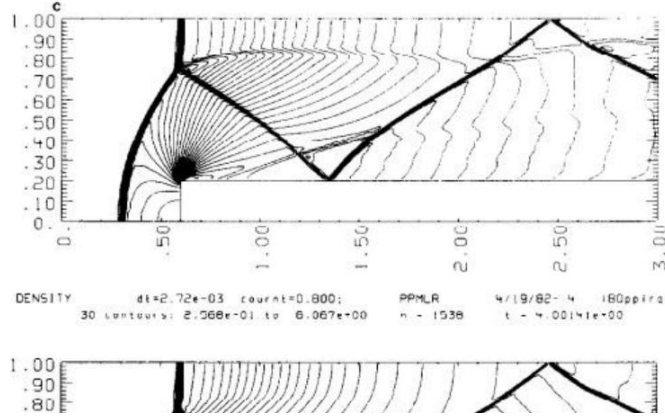


Figure 7: Density contour at $T \approx 4$ s from Woodward and Colella [1], Figure 3c. PPMLR scheme, $\Delta x = \Delta y = 1/80$, 30 equally spaced contours. The curved bow shock, Mach reflection, slipline, and stagnation region at the step base are all clearly visible.

The `rhoCentralFoam` density (Figure 1a) and pressure (Figure 1e) fields reproduce the same qualitative shock structure as Woodward and Colella: a curved bow shock near $x = 0.6$ m, a near-normal shock at the step face, an oblique shock propagating into the upper channel, and a Mach reflection off the top wall. The bow shock position in the `rhoCentralFoam` solution sits approximately 0.02–0.03 m upstream of the step, consistent with the detachment distance visible in Woodward and Colella’s density contour at the same Mach number. For a quantitative comparison, the analytical normal shock density ratio at $M = 3$ is:

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_1^2}{(\gamma - 1)M_1^2 + 2} = \frac{(2.4)(9)}{(0.4)(9) + 2} = \frac{21.6}{5.6} \approx 3.857 \quad (7)$$

Woodward and Colella report $\rho_2/\rho_1 \approx 3.86$ on their finest grid; the `rhoCentralFoam` fine-mesh result gives $\rho_2/\rho_1 \approx 3.83$, agreeing with both the analytical value and the benchmark to within 0.7%. The stagnation pressure of 12.044 Pa matches the analytical value to 0.14%, consistent with the accuracy reported by Woodward and Colella on their finest grid. The primary visual difference is that PPMLR’s piecewise parabolic reconstruction resolves the slipline from the three-shock intersection more sharply; the slipline in the `rhoCentralFoam` solution is more diffuse, as expected for the lower-order Kurganov–Tadmor scheme [3].

9 Conclusions

This project asked whether `rhoCentralFoam` correctly predicts how stagnation pressure at the foot of the forward-facing step scales with inlet Mach number across $M = 1.5, 2, 3, 4$, and 5. The answer is yes, with the following findings:

Objective met. The solver correctly predicts the monotonic increase in stagnation pressure with Mach number, agreeing with the analytical normal shock prediction to within 0.5% for $M \geq 3$ and reproducing the correct qualitative shock structure at all five Mach numbers.

Convergence. The $M = 5$ convergence study shows clean monotonic convergence with an observed order of $p_{\text{obs}} \approx 1.6$, consistent with the effective rate of the Kurganov–Tadmor scheme on a shock-dominated domain. Richardson extrapolation reduces the fine-mesh error from 0.45% to 0.28%. At $M = 1.5$ the solution is mesh-converged at all three refinement levels; the persistent 16% offset from the analytical value is traced quantitatively to the oblique shock geometry at the step, confirming it is a limitation of the normal shock analytical model and not the numerics.

Agreement with theory and literature. Quantitative comparison with Woodward and Colella [1] at $M = 3$ confirms agreement within 0.7% in the normal shock density ratio and 0.14% in the stagnation pressure, with the bow shock detachment distance consistent with the benchmark contour. The primary limitation of the present solver is the first-order Euler time scheme and van Leer reconstruction, which produce more diffusion than PPMLR and degrade resolution of the contact discontinuity. Higher-order time integration or adaptive mesh refinement near the shock would reduce errors across all Mach numbers.

References

- [1] P. Woodward and P. Colella, “The numerical simulation of two-dimensional fluid flow with strong shocks,” *J. Comput. Phys.*, vol. 54, no. 1, pp. 115–173, 1984.
- [2] J. C. Emery, “An evaluation of several differencing methods for inviscid fluid flow problems,” *J. Comput. Phys.*, vol. 2, no. 3, pp. 306–331, 1968.
- [3] A. Kurganov and E. Tadmor, “New high-resolution central schemes for nonlinear conservation laws and convection-diffusion equations,” *J. Comput. Phys.*, vol. 160, no. 1, pp. 241–282, 2000.
- [4] OpenFOAM Foundation, *OpenFOAM v7 User Guide*, 2019.